

● PS E:\Decode AI Internship\Project 2\DecodeLabs_Project2\DecodeLabs_Project2> python main.py
Matplotlib is building the font cache; this may take a moment.

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DecodeLabs | Project 2: Data Classification
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[STEP 1] Loading Iris Dataset...

--- Dataset Overview ---

Total Samples : 150

Total Features : 4

Classes : ['setosa', 'versicolor', 'virginica']

--- First 5 Rows ---

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species	species_name
0	5.1	3.5	1.4	0.2	0	Setosa
1	4.9	3.0	1.4	0.2	0	Setosa
2	4.7	3.2	1.3	0.2	0	Setosa
3	4.6	3.1	1.5	0.2	0	Setosa
4	5.0	3.6	1.4	0.2	0	Setosa

--- Dataset Statistics ---

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
count	150.00	150.00	150.00	150.00	150.00
mean	5.84	3.06	3.76	1.20	1.00
std	0.83	0.44	1.77	0.76	0.82
min	4.30	2.00	1.00	0.10	0.00
25%	5.10	2.80	1.60	0.30	0.00
50%	5.80	3.00	4.35	1.30	1.00
75%	6.40	3.30	5.10	1.80	2.00
max	7.90	4.40	6.90	2.50	2.00

--- Class Distribution ---

```
species_name
Setosa      50
Versicolor  50
Virginica   50
Name: count, dtype: int64
```

[STEP 2] Preparing Features and Target...

Feature Matrix Shape : (150, 4)

Target Vector Shape : (150,)

[STEP 3] Applying Feature Scaling (StandardScaler)...

Before Scaling – Mean of feature 1: 5.843

After Scaling – Mean of feature 1: -0.0

Scaling Done: Mean=0, Variance=1 ✓

[STEP 4] Splitting Data – 80% Train / 20% Test...

Training Samples : 120

Testing Samples : 30

[STEP 5] Finding Optimal K Value...

Best K Found: 2 (Lowest Error Rate)

Graph saved → outputs/k_optimization.png

[STEP 6] Training KNN Model with K=2...

Model Training Complete ✓

[STEP 7] Making Predictions on Test Data...

Sample Predictions vs Actual:

Actual	Predicted	Result
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```

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versicolor    versicolor    ✓ Correct
setosa         setosa         ✓ Correct
virginica      virginica      ✓ Correct
versicolor    versicolor    ✓ Correct
versicolor    versicolor    ✓ Correct
setosa         setosa         ✓ Correct
versicolor    versicolor    ✓ Correct
virginica      virginica      ✓ Correct
versicolor    versicolor    ✓ Correct
versicolor    versicolor    ✓ Correct

```

[STEP 8] Evaluating Model Performance...

Accuracy : 100.00%

F1 Score : 1.0000

--- Classification Report ---

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	10
versicolor	1.00	1.00	1.00	9
virginica	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

Graph saved → outputs/confusion_matrix.png

[STEP 9] Predicting a Brand New Sample...

Input : Sepal=5.1cm x 3.5cm | Petal=1.4cm x 0.2cm

[STEP 9] Predicting a Brand New Sample...

Input : Sepal=5.1cm x 3.5cm | Petal=1.4cm x 0.2cm

Predicted Species → SETOSA 🌸

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PROJECT 2 COMPLETE — DECODELABS

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Dataset : Iris (150 samples, 3 classes)

Algorithm : K-Nearest Neighbors (K=2)

Scaler : StandardScaler

Train/Test : 80% / 20%

Accuracy : 100.00%

F1 Score : 1.0000

Outputs saved : outputs/

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